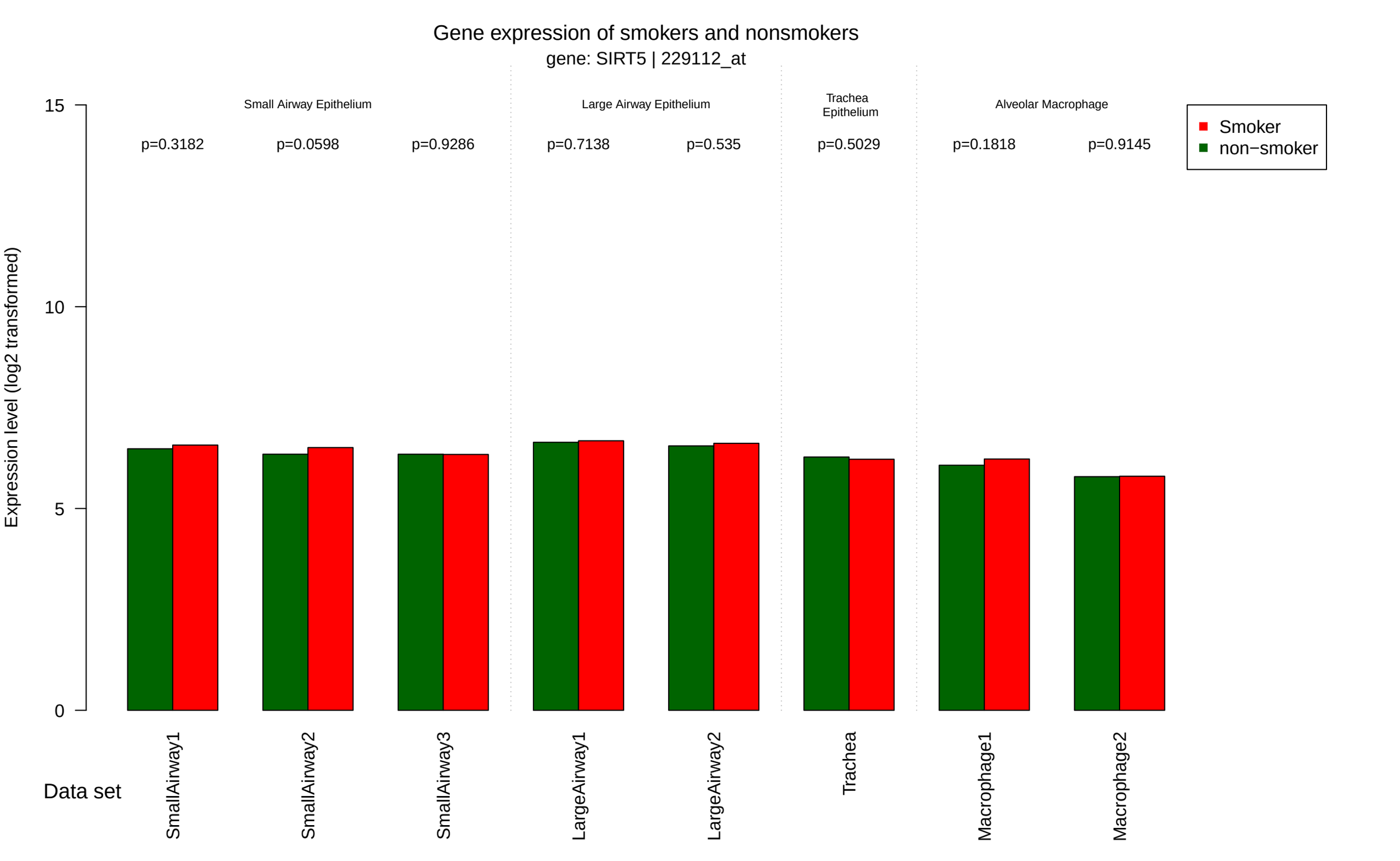
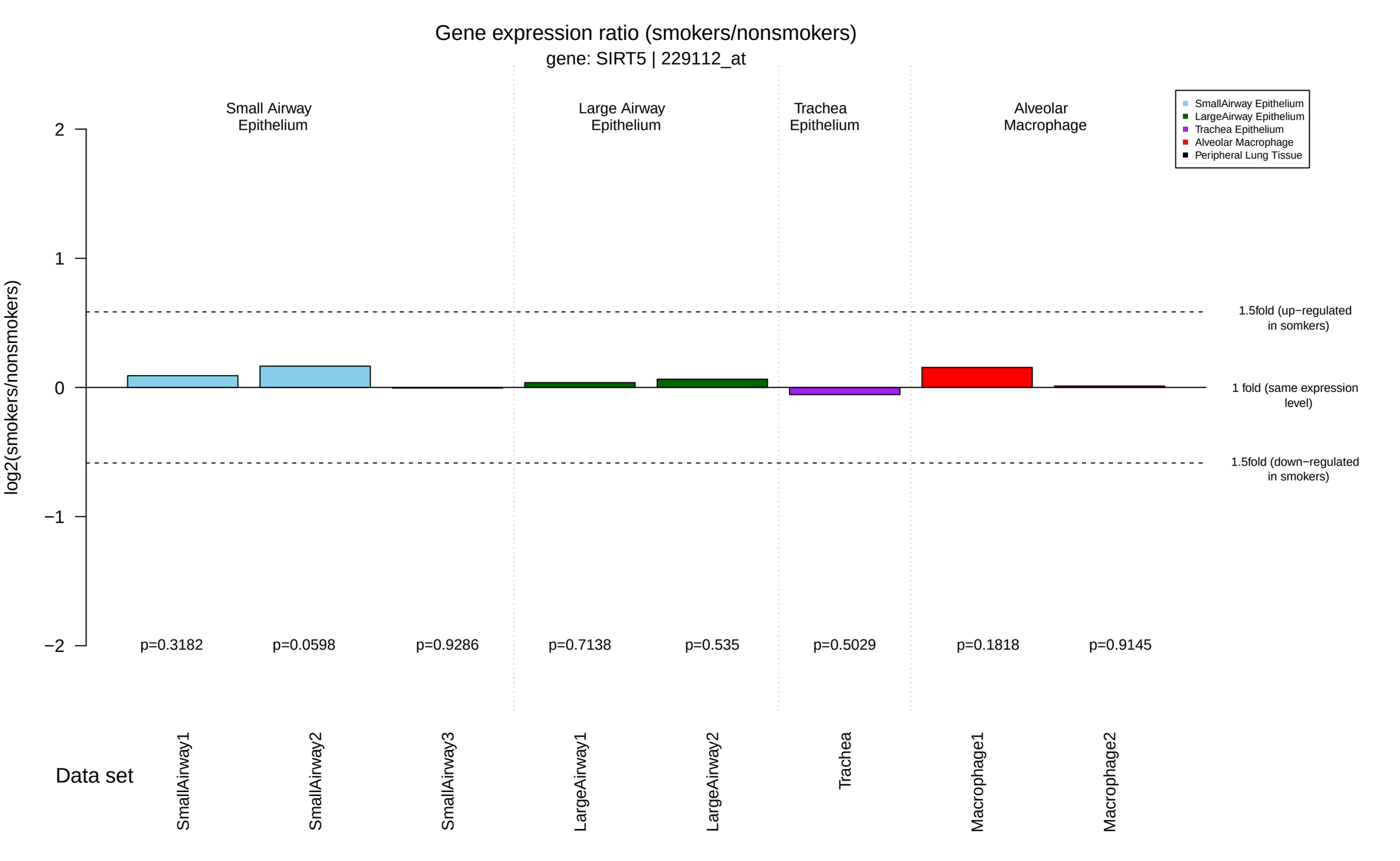
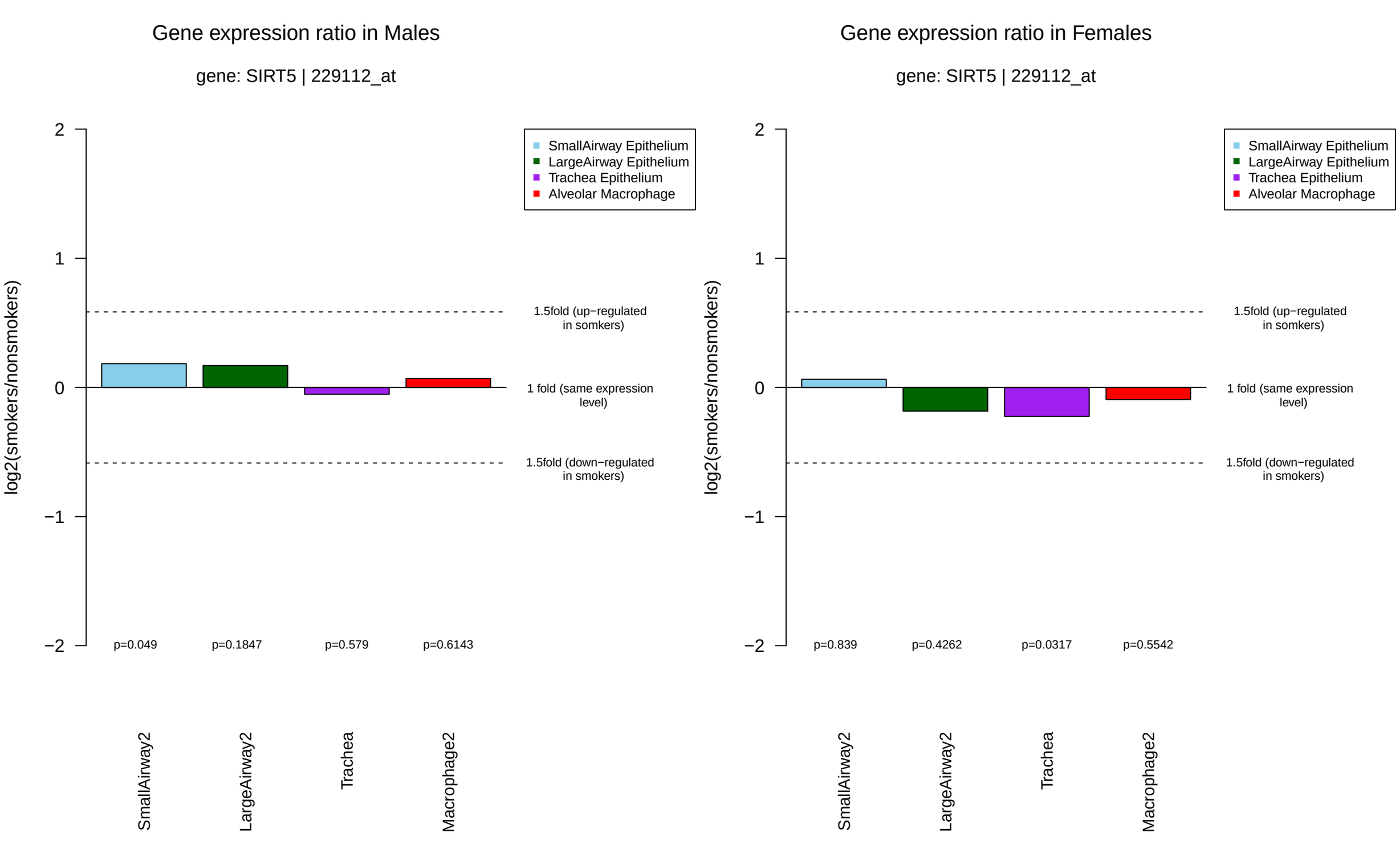
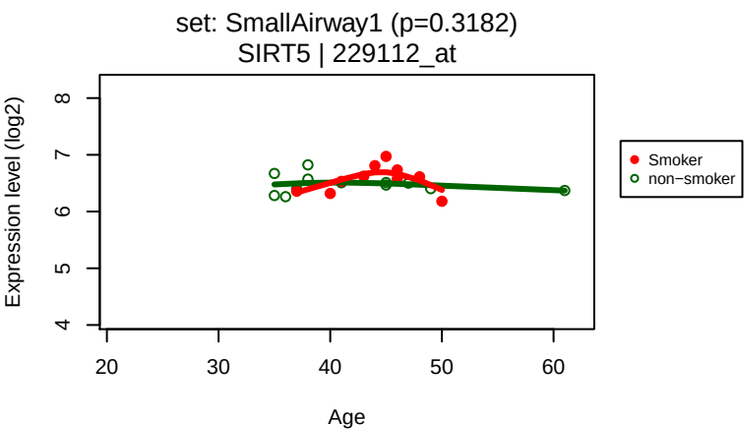
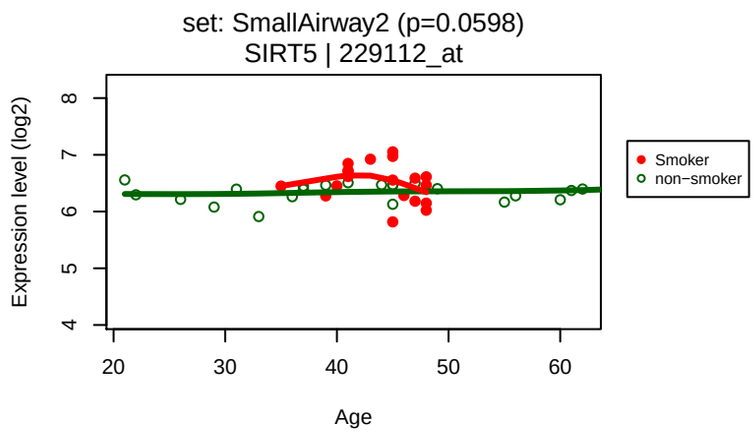
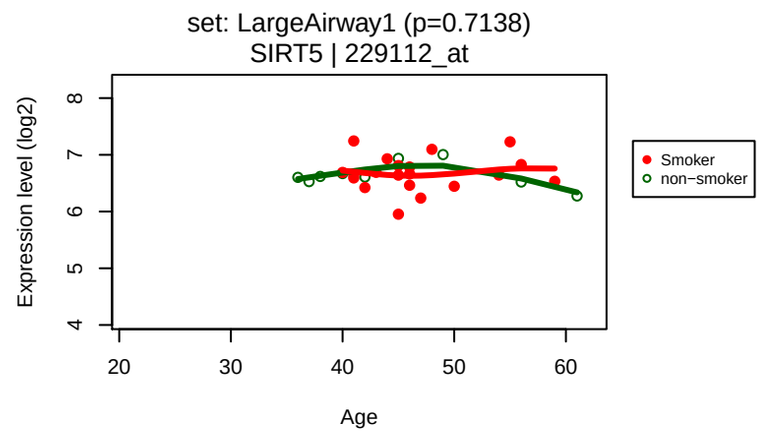

Gene expression of smokers and nonsmokers
gene: SIRT5 | 229112_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3182
p=0.0598
p=0.9286
p=0.7138
p=0.535
p=0.5029
p=0.1818
p=0.9145
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

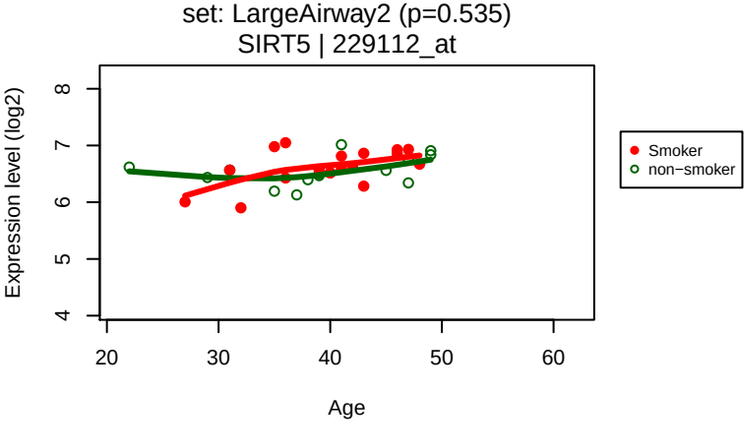

Gene expression ratio (smokers/nonsmokers)
gene: SIRT5 | 229112_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3182
p=0.0598
p=0.9286
p=0.7138
p=0.535
p=0.5029
p=0.1818
p=0.9145
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

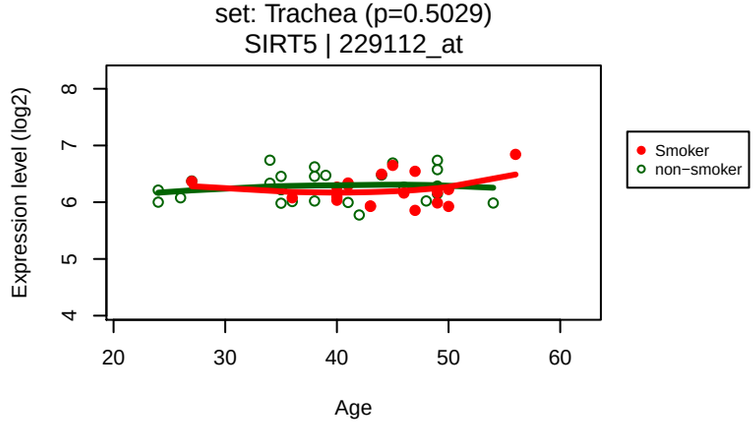

Gene expression ratio in Males
gene: SIRT5 | 229112_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.049
p=0.1847
p=0.579
p=0.6143
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIRT5 | 229112_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.839
p=0.4262
p=0.0317
p=0.5542
SmallAirway2
LargeAirway2
Trachea
Macrophage2

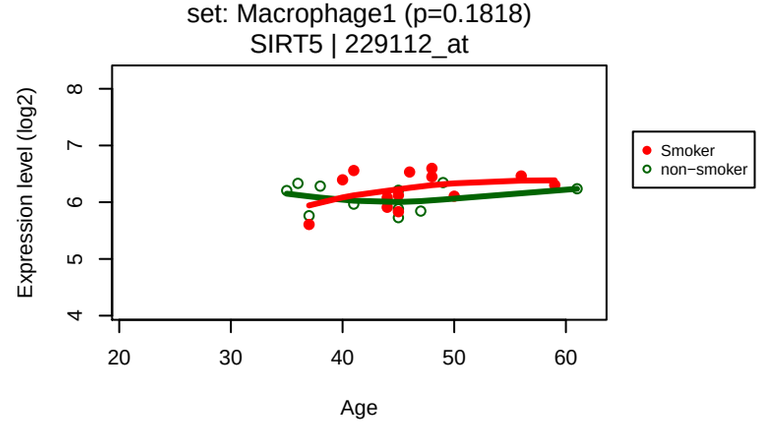

set: SmallAirway1 (p=0.3182)
SIRT5 | 229112_at
Expression level (log2)
Age
Smoker
non−smoker

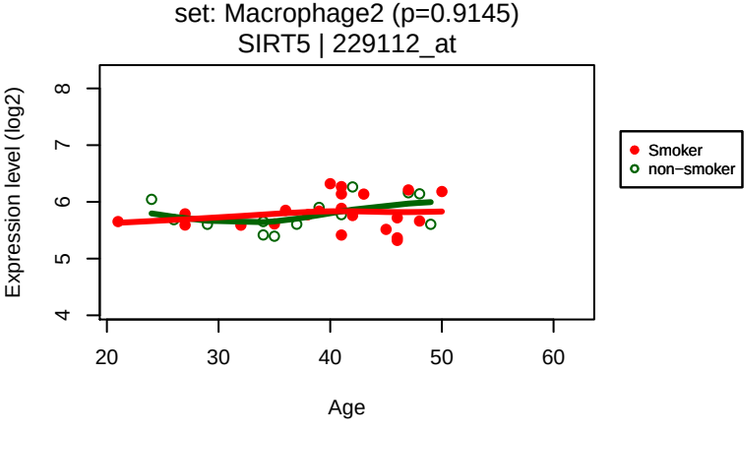

set: SmallAirway2 (p=0.0598)
SIRT5 | 229112_at
Expression level (log2)
Age
Smoker
non−smoker

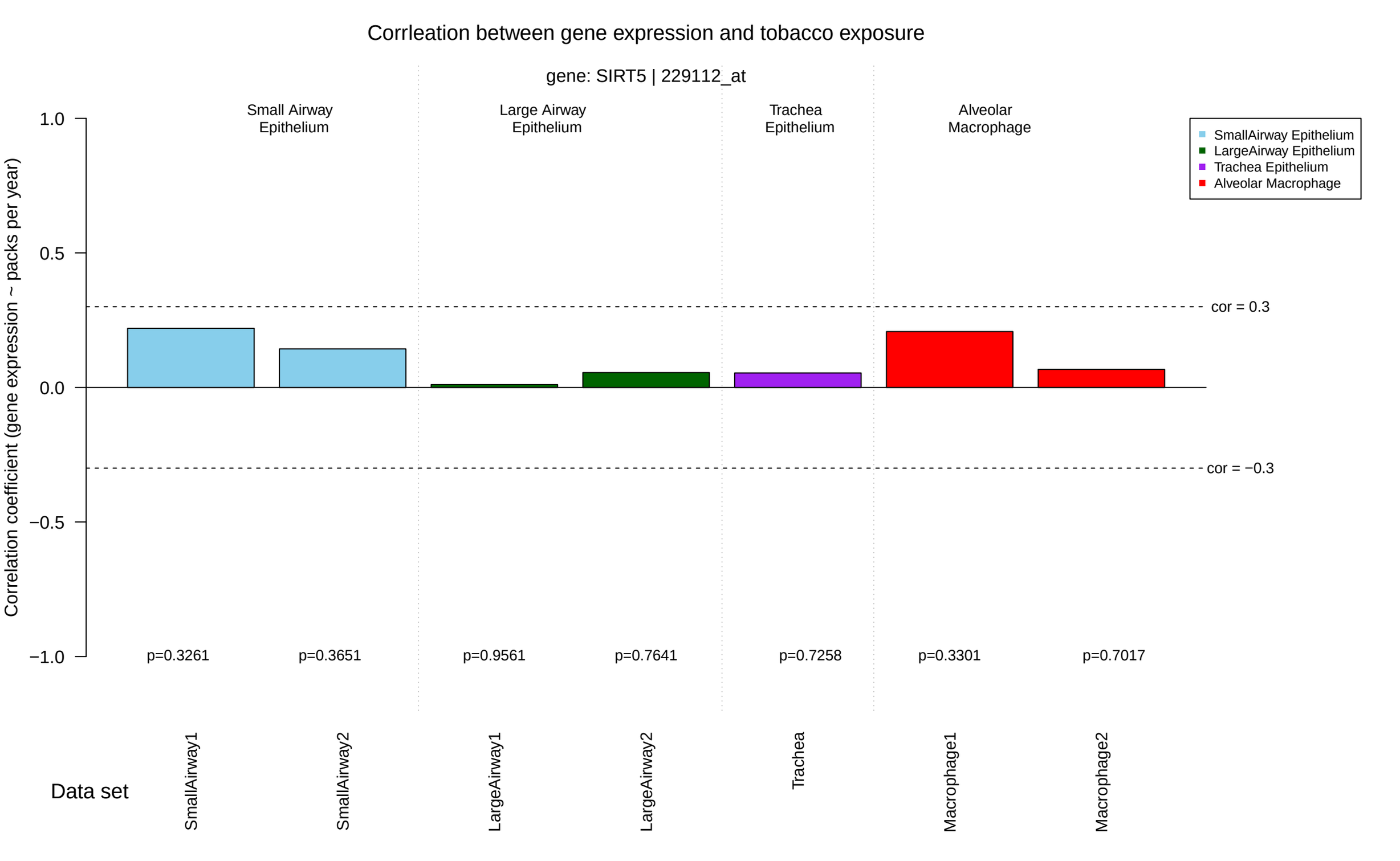

set: LargeAirway1 (p=0.7138)
SIRT5 | 229112_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.535)
SIRT5 | 229112_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5029)
SIRT5 | 229112_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1818)
SIRT5 | 229112_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9145)
SIRT5 | 229112_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIRT5 | 229112_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3261
p=0.3651
p=0.9561
p=0.7641
p=0.7258
p=0.3301
p=0.7017
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2